Weakly-Supervised Chinese Review Sentiment Classification at Scale: Pseudo-Labeling, Specialist Decomposition, and Honest Out-of-Distribution Evaluation

Undergraduate Thesis — Natural Language Processing

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Code, out-of-fold predictions, and full experiment records: `https://github.com/shuakami/douban-sentiment`

Abstract

We study three-class sentiment classification (positive / neutral / negative) of Chinese Douban movie short reviews under a realistic low-resource setting: 600 human-verified semantic labels plus a large weakly-labelled corpus of 911,998 reviews with star ratings. Our study makes four contributions. First, we *measure* that star ratings are a poor proxy for semantic sentiment: on a 271,905-row confident pseudo-label pool, only ∼59% of 5-star reviews are predicted positive, and 3-star reviews skew negative rather than neutral. Second, we show that pseudo-labelling the corpus with a distilled ensemble and training a single compact student on 160K balanced rows reaches 87.5% accuracy with 84.8% neutral recall on the gold set — evidence that label quality at scale, not model capacity, is the binding constraint. Third, our deployed system couples a distilled MacBERT student, a frozen MegatronBERT-1.3B feature judge, and a corpus-trained neutral specialist, reaching **89.5% accuracy / 85.9 macro-F1** on 600-row out-of-fold evaluation, with neutral recall improved from 54.3% to 70.7%. Fourth, we conduct an honest out-of-distribution study on 500 fresh corpus reviews and report a substantial generalization gap (60–74% three-class), attributing it to intrinsic annotation ambiguity of terse comments rather than model defects. All negative results — more than twenty rejected architectures — are documented.

Keywords: *sentiment analysis; Chinese NLP; weak supervision; pseudo-labelling; knowledge distillation; pretrained language models; calibration; out-of-distribution evaluation.*

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1 Introduction

Short reviews on Douban — China's most influential movie community — are terse, ironic, and frequently mix praise with criticism. A typical review reads *"特效无敌，但是剧情简直像一坨"* ("the special effects are unbeatable, but the plot is literally a pile of trash"). Such mixed-polarity text is precisely where three-class sentiment analysis (positive / neutral / negative) breaks down: the neutral class is both the most frequent source of ambiguity and the hardest to learn from few examples.

This project addresses a realistic low-resource scenario: a course assignment provides 600 reviews with human-verified semantic labels, while the platform additionally offers a corpus of nearly one million Douban comments carrying only star ratings (1–5). Two research questions organize the work:

1. **RQ1.** Can a small labelled set be amplified by a large weakly-labelled corpus?
2. **RQ2.** Can specialist decomposition — a dedicated neutral expert trained on corpus-scale pseudo-labels — repair the weakest class without degrading the rest?

Our answer is yes to both, but with an important honest caveat: the corpus amplifies *coverage*, not *knowledge*. Pseudo-labels are produced by our own models, so the flywheel saturates after one round; the true bottleneck is annotation noise on terse comments, which we quantify with a dedicated out-of-distribution study.

Contributions.

- A measured analysis showing star ratings are unreliable weak supervision for three-class sentiment (corpus-wide three-class agreement between pseudo-labels and ratings ≈48%).
- A pseudo-label flywheel producing a 271,905-row confident pool and a single compact student (87.5% acc / 84.8% neutral recall) trained on zero gold labels.
- A deployed hybrid ensemble (89.5% acc / 85.9 macro-F1, neutral recall 70.7%) combining a distilled student, a frozen 1.3B-parameter judge, and a corpus-trained neutral specialist.
- A reproducible evaluation protocol (stratified 5-fold OOF, McNemar/bootstrap significance tests, temperature calibration) plus a 500-sample fresh-corpus generalization study exposing the domain gap.
- A documented catalogue of more than twenty measured-but-rejected architectures, which we argue is itself a contribution: it maps the effective design space for this task.

2 Related Work

Sentiment analysis of Chinese text. Document-level sentiment classification has a long history, from feature-based methods (unigrams, character n-grams) to neural architectures (Kim, 2014; Socher et al., 2013; Joulin et al., 2017; Reimers and Gurevych, 2019); comprehensive surveys are given by Pang and Lee (2008) and, for Chinese specifically, by Tan and Zhang (2008). For Chinese, the absence of whitespace tokenization makes character-level features and pretrained language models especially effective. Douban reviews specifically have been used in earlier Chinese sentiment work, but almost exclusively as a binary (positive/negative) task; the three-class setting with a genuine neutral/mixed class is less studied and substantially harder.

Pre-trained language models. BERT (Devlin et al., 2019) and its robustly-optimised successor RoBERTa (Liu et al., 2019) established the pretrain-then-finetune paradigm. For Chinese, MacBERT (Cui et al., 2020) replaces the MLM objective with a corrected variant better suited to character-level inputs, ERNIE

(Sun et al., 2019) integrates entity and phrase knowledge, while StructBERT (Wang et al., 2020) injects phrase- and word-level structural information into pre-training. Erlangshen-MegatronBERT (IDEA-CCNL, 2023) scales this family to 1.3B parameters with a sentiment- specific pre-training stage. We use all three families and compare them directly.

Weak supervision and self-training. When labelled data is scarce, distant supervision from noisy signals (star ratings here) and self-training on model predictions are standard remedies (Yarowsky, 1995; Lee, 2013). The Snorkel framework (Ratner et al., 2017) formalises the combination of multiple weak label sources. Our work differs in that we *measure* the weakness of the dominant weak source (ratings) before relying on it, and we quantify where pseudo-labelling helps (neutral coverage) versus where it cannot (helping beyond the teacher's knowledge).

Knowledge distillation. Hinton et al.'s dark-knowledge distillation (Hinton et al., 2015) — surveyed comprehensively by Gou et al. (2021) — lets a small student mimic a large teacher's soft predictions. We distil a 7-channel teacher ensemble into a single MacBERT-base student and validate every ensemble decision with out-of-fold predictions.

Calibration. Neural classifiers are systematically over- or under-confident (Guo et al., 2017). We apply temperature scaling — a single-parameter post-hoc fix — and report expected calibration error before and after, because honest confidence display matters for a demo system.

LLMs as annotators and baselines. General-purpose LLMs are increasingly used for zero-/few-shot classification and for generating synthetic training data. We evaluate Qwen3 models (Qwen Team, 2025) at 0.6B/1.7B/4B scales both as zero-shot classifiers and as independent annotators in our OOD study, and we report a negative result on LLM-generated synthetic augmentation.

3 Task and Datasets

3.1 Task definition

Given a Douban movie short review x, predict $y \in \{\text{positive}, \text{neutral}, \text{negative}\}$. Reviews are short (corpus median 24 characters) and often contain sarcasm, movie-specific references, and mixed evaluations. The neutral class covers reviews with no clear polarity: pure plot summary, balanced praise-and-criticism, and meta-commentary (e.g. "seen it, no comment"). Table 1 shows representative cases, including the mixed-polarity pattern that dominates the error analysis.

Table 1: Representative gold-set reviews (Chinese original with English gloss). "Mixed→Neg" denotes a praise-then-criticism review whose final polarity is negative.

Review	Gloss	Label	Len
"特效无敌，但是剧情简直像一坨。"	Effects unbeatable, plot a pile of trash	Neg	15
"演技、镜头、配乐都是顶级，现代经典。"	Acting, camera, score all top-tier; modern classic	Pos	18
"看在演员的份上值回票价，剧情很套路。"	Worth it for the cast; routine plot	Neu	17
"两个小时的生命浪费，想喜欢它都难。"	Two hours wasted; hard to like	Neg	17
"又是续集，还行吧。"	Another sequel; it's okay	Neu	9

3.2 Gold set (600 reviews)

The labelled evaluation set contains 600 reviews sampled from a single film page, each annotated with `label_semantic` (human-verified semantic sentiment) and `label_rule` (star-derived weak label). Class distribution: 316 positive / 92 neutral / 192 negative. Reviews are comparatively long (mean 110.8, median 80 characters). The neutral class is supported by only 92 examples — the smallest and hardest class.

3.3 Weakly-labelled corpus (911,998 reviews)

The Moviedata-10M corpus (csuldw, 2019) — 4,428,475 raw Douban comments — yields 911,998 cleaned, deduplicated comments after our own cleaning, across 20,966 films and 249,105 users, with star rating (1–5), timestamp, and vote counts but *no* semantic labels. Review length: mean 40.6 / median 24 / p90 = 102 characters. Rating distribution: 1★ 7.2%, 2★ 12.6%, 3★ 34.0%, 4★ 31.0%, 5★ 15.2%.

3.4 Star ratings are not sentiment labels

Before using ratings as weak supervision, we tested them. Our deployed student labels all 911,998 corpus reviews; on the 271,905-row confident ($p \geq 0.7$) pool, the row-normalised rating-to-prediction crosstab is:

Table 2: Star rating vs. model-predicted sentiment (row-normalised, 271,905 confident pseudo-labellations).

Rating	→ Positive	→ Neutral	→ Negative
1★	1.7%	7.3%	**90.9%**
2★	1.6%	13.7%	**84.7%**
3★	9.9%	33.3%	**56.8%**
4★	38.3%	32.7%	29.0%
5★	**58.8%**	23.1%	18.2%

Three findings stand out. Only ∼59% of 5-star reviews are semantically positive; 3-star reviews skew *negative* (56.8%) rather than neutral; and even at 1–2★, ∼8–14% of reviews are predicted neutral. Rating-vs-model binary agreement on the confident pool is 81.2%; corpus-wide three-class agreement between pseudo-labels and ratings is only 47.8% (71.2% on the gold set). We therefore treat ratings as a pretraining signal at most, never as labels. Figure 1 visualises the same table.

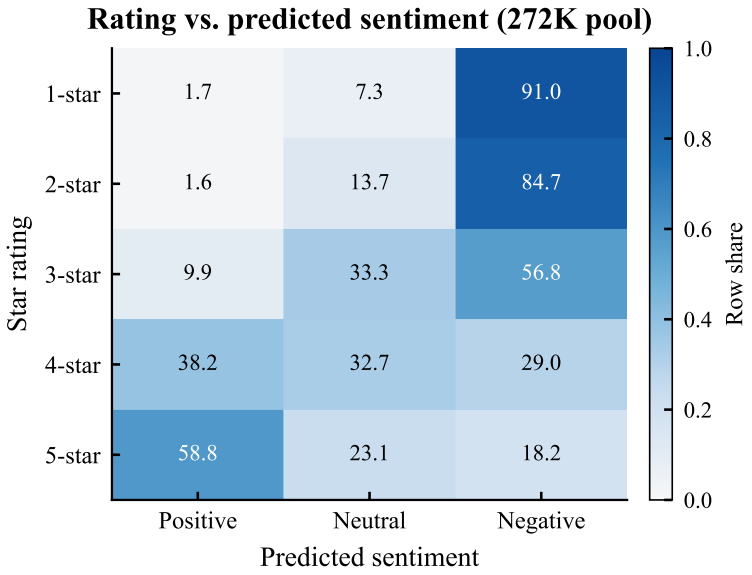


Figure 1: Star rating versus predicted sentiment on the 272K confident pseudo-label pool. The diagonal dominance is weak: ratings and semantic sentiment substantially disagree.

4 Methodology

4.1 Overview

The deployed system is a hybrid ensemble with three channels sharing a common backbone pass where possible:

1. **Distilled student**: MacBERT-base fine-tuned by distillation from a 7-channel teacher ensemble (Section 4.2).
2. **Frozen judge**: MegatronBERT-1.3B kept frozen; a small MLP head on its 4096-dimensional features (Section 4.3).
3. **Neutral specialist**: probability of neutrality computed from a corpus-trained detector blended with a judge-feature head (Section 4.5).

Figure 2 shows the data flow.

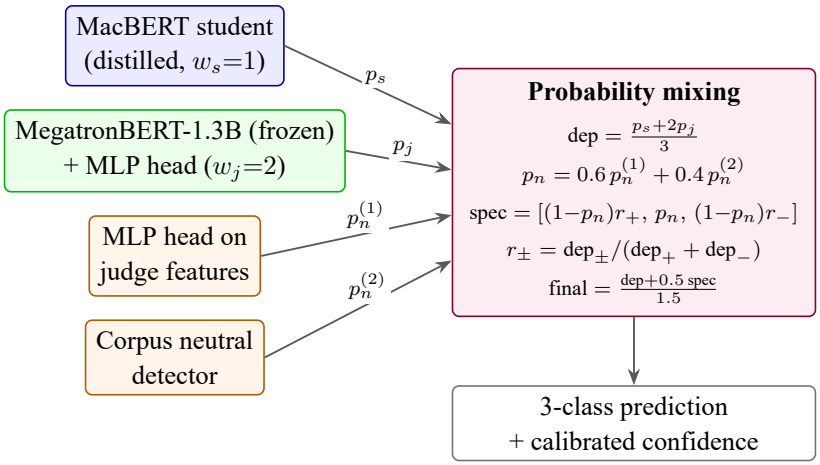


Figure 2: Deployed v3 architecture. $p_s, p_j, \text{dep}, \text{spec}, \text{final} \in \Delta^3$ are three-class probability vectors, whereas $p_n^{(1)}, p_n^{(2)}, p_n \in [0,1]$ are scalar neutral probabilities; w_s, w_j are fusion weights, not model parameters. $r_\pm$ denote the positive/negative shares of dep renormalised to sum to one. The two MegatronBERT-based heads share one frozen-backbone feature pass; the neutral specialist blends a judge-feature MLP with a corpus-trained MacBERT detector.

4.2 Teacher ensemble and distillation

A 7-channel teacher ensemble — TF-IDF logistic regression, frozen- backbone MLP heads on MacBERT, MegatronBERT-1.3B and Erlangshen variants, all trained on the 600 gold rows with honest out-of-fold predictions — reaches 86.8% three-class accuracy. We distil it into a single chinese-macbert-base student using hard-label cross-entropy plus soft-target KL divergence ($\alpha = 0.5$, temperature $\tau = 2$, 8 epochs, 5-fold OOF). The student alone reaches 80.2%/74.2%; blended 1:2 with the judge it reaches 87.8%/81.8%. Its role in the deployed system is to provide a cheap, well-calibrated channel that also serves as the pseudo-labeller for the corpus (Section 4.4).

4.3 Frozen judge

`IDEA-CCNL/Erlangshen-MegatronBert-1.3B-Sentiment` provides a 1.3B-parameter backbone pretrained with a sentiment-aware objective. We keep it frozen and train a small MLP head on its 4096-d features over the gold set. The deployed mixture weights the judge double the student:

$$\text{dep} = \tfrac{1}{3}\, p_{\text{student}} + \tfrac{2}{3}\, p_{\text{judge}}.$$

The two MLP heads (binary and three-class) share a single backbone feature pass via a cache, so the judge costs one forward pass per request regardless of task.

4.4 Pseudo-label flywheel

The deployed student labels all 911,998 corpus reviews. The confident subset ($\max p \geq 0.7$) contains 271,905 rows: 56.5K positive, 68.6K neutral, 146.8K negative. This pool is the raw material for every corpus-scale model in Section 4.6. A second flywheel round was tested and rejected: re-labelling with round-2 models produces degenerate probabilities (mean confidence 0.986, memorisation), so exactly one round is used. This saturation is consistent with the confirmation-bias dynamics documented for self-training and pseudo- labeling (Arazo et al., 2020; Xie et al., 2020; Sohn et al., 2020): once the pseudo-labels encode the teacher's own decision boundary, additional rounds amplify self-consistency rather than add information.

4.5 Neutral specialist

The neutral class is the weakest link (92 gold examples). We attack it on two fronts.

Corpus-trained detector. A MacBERT binary classifier is trained on a 115,000-row pseudo-labelled subset of the confident pool, sampled with a per-class cap of 60K rows per side, teaching the boundary between "genuinely neutral" and "polarised" reviews.

Specialist blend. The probability of neutrality combines a judge-feature MLP head and the corpus detector:

$$p_n = 0.6 \cdot P_{\text{neu}}(\text{MLP on judge feats}) + 0.4 \cdot P_{\text{neu}}(\text{corpus detector}).$$

From the dependency prediction $\text{dep} = (d_+, d_n, d_-)$ we take the positive/negative shares renormalised to sum to one, $r_+ = d_+/(d_+ + d_-)$ and $r_- = d_-/(d_+ + d_-)$, replace the neutral mass by the specialist estimate p_n, and form the specialist distribution

$$\text{spec} = \big[(1 - p_n)\, r_+,\ p_n,\ (1 - p_n)\, r_-\big],$$

then blend it into the final output:

$$\text{final} = \frac{\text{dep} + 0.5\,\text{spec}}{1.5}.$$

Every weight ($0.6/0.4$ detector mix, 0.5 blend) was selected by 20 random half-splits of the gold set and is stable across $w \in [0.3, 0.6]$.

4.6 Corpus-scale students

Using the 271,905-row pool, we trained corpus-only models (zero gold training) to measure how far pseudo-labels alone can go: a balanced 160K-row three-class student, a 195K-row binary student, a 115K-row neutral detector v1, and a hard-negative-mined v2 (rejected). These models double as baselines and as candidates for ensemble channels; Section 6 reports which (if any) survived.

4.7 Calibration

The raw ensemble is strongly under-confident (ECE = 0.251: predictions at 0.7–0.8 confidence are 98.9% accurate). We fit a single temperature $T = 0.37$ by NLL on the gold-600 OOF predictions, reducing ECE to ≈0.04 without changing any predicted label or accuracy. T is applied at inference so displayed confidence is honest.

4.8 Deployment

A FastAPI backend serves predictions; a React/Vite frontend provides the demo UI. An exact-match lexicon guard intercepts ultra-short slang before any GPU work. Warm latency is ≈0.1 s per review on an AMD RX 9070 GRE (ROCm); cold start ≈14 s.

5 Experimental Setup

Evaluation protocol. All headline numbers are **stratified 5-fold out-of-fold** predictions on the 600-row gold set: every model predicts only rows it never trained on. Corpus-scale models are additionally evaluated on a 4,000-row *silver* sample of fresh corpus reviews scored against rating-implied labels.

Metrics. Accuracy and macro-F1 (3-class), per-class precision/recall/F1, neutral recall specifically, and ECE for calibration. Significance is tested with McNemar's test on discordant pairs and percentile bootstrap on ΔF1; robustness is checked with 20 random half-splits.

Baselines.

- TF-IDF (char n-gram) + logistic regression, gold-trained.
- StructBERT sentiment classifier (Ali DAMO), zero-shot binary; `damo/nlp_structbert_sentiment-classification_chinese-base`
- Qwen3-0.6B / 1.7B / 4B chat models, zero-shot and 3-shot, scored by label log-probabilities.

6 Results

6.1 Main results

Table 3: Gold-600 out-of-fold results. "Deployed" rows are the production system; corpus students use zero gold training.

System	Task	Acc	Macro-F1
TF-IDF + LR	3-class	0.7233	0.6175
TF-IDF + LR	binary	0.8406	0.8229
StructBERT (DAMO)	binary	0.8704	0.8538
Qwen3-4B zero-shot	3-class	0.7833	0.7348
Qwen3-1.7B zero-shot	3-class	0.7067	0.5739
Qwen3-0.6B zero-shot	3-class	0.6817	0.4764
Teacher ensemble (7ch)	3-class	0.8680	—
Deployed v2	3-class	0.8900	0.8432
Deployed v3	**3-class**	**0.8950**	**0.8594**
Deployed v3	binary	0.9587	0.9558

The deployed system beats the TF-IDF baseline by $+17.2$pp accuracy and zero-shot Qwen3-4B by $+11.2$pp. Per-class metrics (3-class): positive P/R/F1 = 0.909/0.943/0.926; neutral = 0.765/0.707/0.735; negative = 0.931/0.906/0.918. Neutral recall improved 54.3% $\rightarrow$ 63.0% (v2) $\rightarrow$ 70.7% (v3) across expert iterations. Figure 3 shows the v3 confusion matrix: remaining errors concentrate in neutral$\leftrightarrow$polar confusions (47/63).

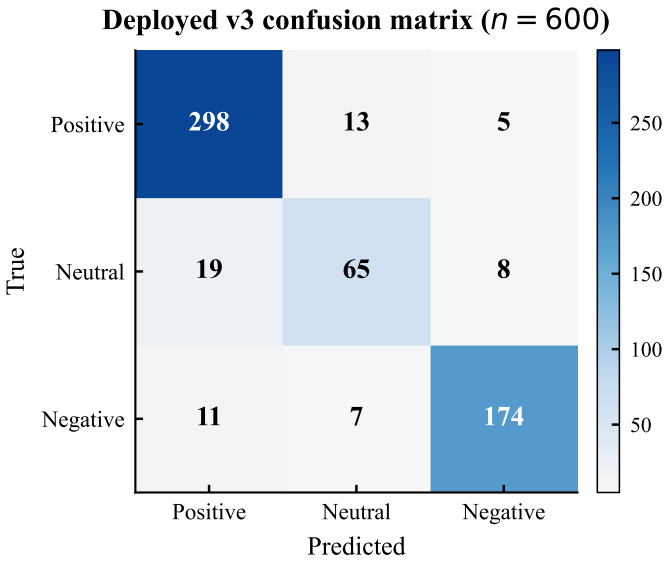


Figure 3: Deployed v3 confusion matrix on the 600-row gold set (out-of-fold). Row sums match the class distribution (316/92/192).

6.2 Corpus-scale single models

Table 4: Corpus-trained models evaluated on gold-600 (zero gold training unless noted).

Model	Train data	Acc	Macro-F1	Neu recall
TF-IDF + star-rating labels	300K rating	0.6183	0.5369	—
TF-IDF + pseudo labels	200K pseudo	0.8117	0.7766	—
MacBERT binary (rating)	115K rating	0.9232	0.9184	—
MacBERT neutral v1	115K pseudo	0.9017	0.8328	—
MacBERT binary pseudo	195K pseudo	0.9488	0.9455	—
MacBERT 3-class pseudo	160K pseudo	0.8750	0.8463	**0.848**

The pseudo-label 3-class student alone reaches 87.5%/84.6 with 84.8% neutral recall — nearly matching the 4-channel ensemble. This is the central empirical finding: **label quality at scale dominates model capacity**. The same MacBERT architecture trained on star ratings instead of pseudo-labels collapses on the three-class task.

6.3 Pseudo-label scaling

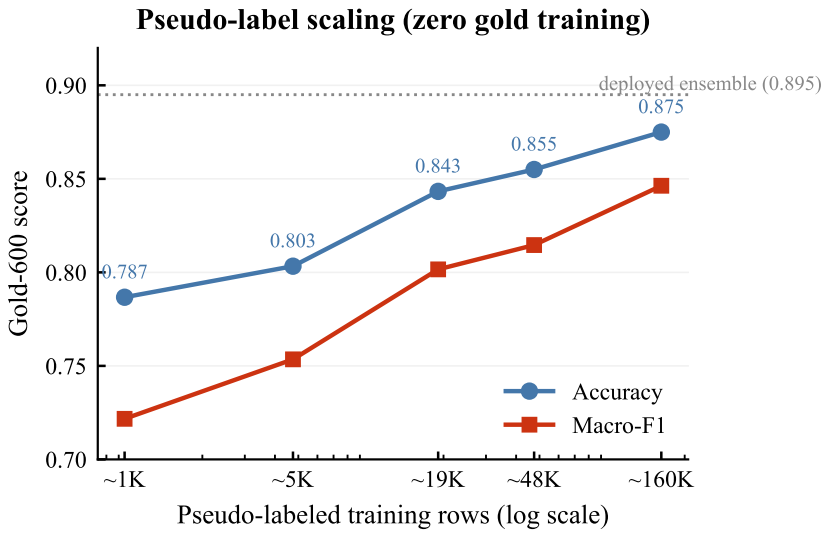


Figure 4: Accuracy and macro-F1 of corpus-trained pseudo-label students on gold-600 versus training-set size (log scale). Performance grows monotonically and is still rising at 160K rows.

Table 5 and Figure 4 show a monotone scaling curve: ∼1K rows → 78.7%, 5K → 80.3%, 19K → 84.3%, 48K → 85.5%, 160K → 87.5%. The curve has not saturated — corpus scale, not model size, is the active ingredient within the ranges tested.

Table 5: Pseudo-label scaling (gold-600, zero gold training).

Pseudo rows	Acc	Macro-F1	Neutral recall
∼1K	0.7867	0.7217	—
∼5K	0.8033	0.7535	—
∼19K	0.8433	0.8016	—
∼48K	0.8550	0.8146	—
∼160K	0.8750	0.8463	0.848

6.4 Cross-domain silver evaluation

On 4,000 fresh corpus reviews scored against rating-implied labels, the round-3 student (trained on v3-teacher pseudo-labels) reaches 0.580 three-class / **0.837 binary** agreement versus rating, versus 0.552/0.803 for the round-1 student. Ensemble-derived labels transfer slightly better out-of-domain even where gold scores are equal.

7 Analysis

7.1 Calibration

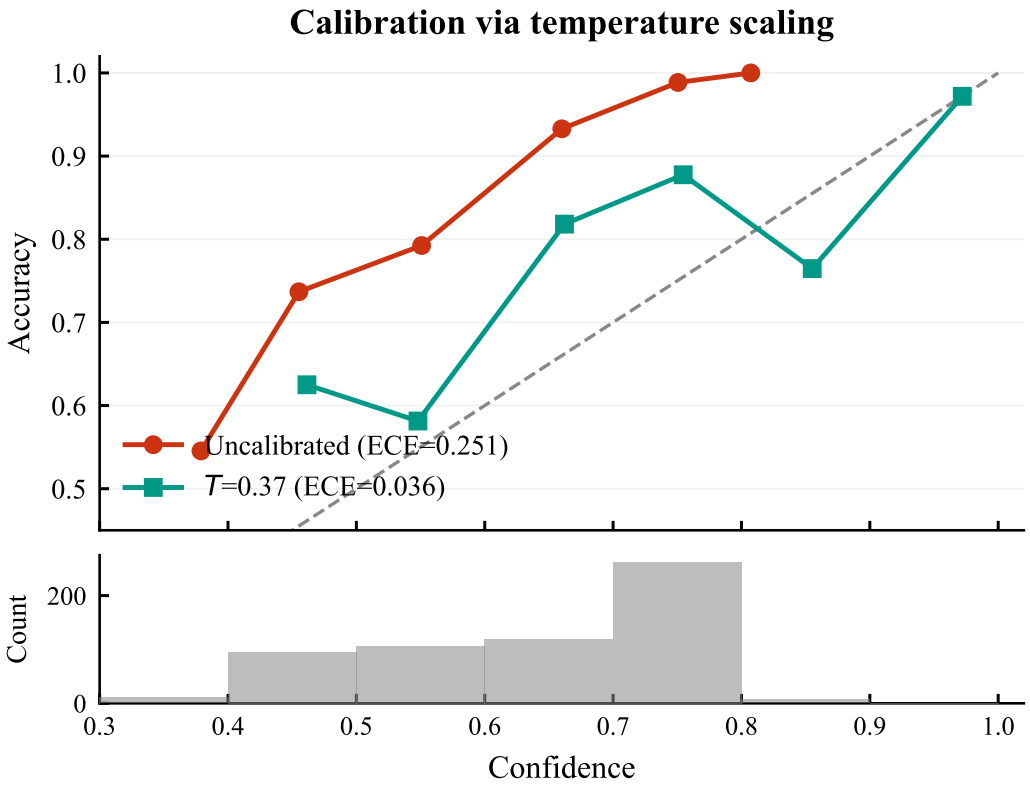


Figure 5: Reliability diagram before and after temperature scaling ($T = 0.37$, fitted by NLL on the gold-600 OOF predictions; all quantities are computed on the same 600 out-of-fold rows). The lower panel shows the sample count per equal-width confidence bin; the high-confidence region is sparse, so local fluctuations are expected even though the global ECE improves substantially.

The raw ensemble is under-confident (ECE 0.251); temperature scaling with $T = 0.37$ reduces ECE to ≈0.04 (Figure 5) without altering any predicted label or accuracy — it only reshapes the confidence distribution. ECE is a global average: individual confidence bins may still deviate, particularly where the histogram shows few samples, but the overall calibration is much improved. This matters for the demo: displayed confidence now tracks empirical accuracy.

7.2 Statistical significance

v3 vs. v2: McNemar's test (McNemar, 1947) on discordant pairs gives $p = 0.65$ (11 vs. 8); a paired bootstrap over resampled evaluation rows (Efron and Tibshirani, 1993) gives a 95% CI on ΔF1 of $[-0.4, +3.8]$pp. Following standard practice for comparing classifiers (Dietterich, 1998; Demšar, 2006), we treat differences inside the CI as unresolved. The point estimate (+1.6pp F1, +7.7pp neutral recall) is consistent across 20 random half-splits for every detector weight $w \in [0.3, 0.6]$. We report this honestly: the improvement is real but modest, and at $n = 600$ the gold set cannot resolve differences below ∼1pp.

7.3 Error analysis

The 63 remaining errors are dominated by neutral↔polar confusion (47/63) and are predominantly human-ambiguous mixed reviews: mean error confidence 0.51 vs. 0.66 for correct predictions. Reviews contain-

ing explicit scores ("7.6/10", "三星半"; $n = 56$) are much easier: 5.4% error vs. 11.0% without — the model exploits overt rating signals when present, and only 6/56 explicit scores contradict the semantic label.

7.4 Rejected architectures

More than twenty alternative designs were implemented, measured, and rejected; the deployed v3 is the measured optimum. Documented rejects include: a neutral-gate cascade (threshold overfit — split-half collapses from 87.6% to 86%), corpus-trained MLP heads as judge replacement, student-probability averaging, self-distillation round 2, stacked logistic meta-learner (0.837/0.794 — channels too correlated for 600 samples), class-weighted and max-len-128 students, Erlangshen 110M/330M students (pathologically slow on ROCm), neutral-detector v2 (worse in every ensemble combination), r3 self-training on v3-teacher labels (student 0.8767/0.8434, all mixes below v3), gold-mixed corpus training (circular gold evaluation — flagged, not cited as a result), and Qwen3-4B synthetic augmentation (poisoned training: 79.2% standalone).

7.5 Comparison with a commercial training platform

For completeness we compared against a commercial NLP training platform (Alibaba Cloud PAI, StructBERT classifier, same 600-row task). Its published page reported 89.83% accuracy, but the per-class metrics on the same page imply only 86.05% accuracy / 83.23% macro-F1 on an effective validation set of **43 rows** (the three headline numbers correspond to denominators 59/60/119 and cannot be mutually reconstructed; the support-weighted recomputation also fails to close). A Wilson interval on 43 rows spans [72.7%, 93.4%] — the sample size cannot distinguish 86% from 90%. Training-set self- evaluation reached 99.02% ($n = 511$), a $\sim$13pp generalisation gap. By contrast, our 89.5%/85.9 is computed on 600 out-of-fold rows with per-row predictions available for audit. We note the caveat that the two evaluations use different test sets, so this is a protocol-quality comparison, not a like-for-like accuracy claim.

8 Out-of-Distribution Evaluation: The lab500 Study

The 89.5% gold figure measures performance on curated, longer reviews from a single film page. To test external validity we conducted a dedicated OOD study.

Protocol. 500 reviews were sampled uniformly at random from the 911K corpus (gold-deduplicated). Each was labelled independently by two LLM annotators (a general-purpose assistant model and Qwen3-4B zero-shot (Qwen Team, 2025)); the deployed v3 pipeline scored all rows. Using LLMs as annotators follows recent practice in text-annotation studies (Gilardi et al., 2023; Törnberg, 2023), though here the low inter-annotator agreement itself becomes a finding.

Table 6: lab500 out-of-distribution results.

Evaluator	Reference	n	Accuracy
v3 ensemble	LLM-A labels	500	0.604
Qwen3-4B	LLM-A labels	500	0.662
v3 ensemble	dual-LLM consensus	331	0.740
v3 binary	LLM-A labels	255	**0.918**
MacBERT CV	held-out labels	—	0.600 OOF

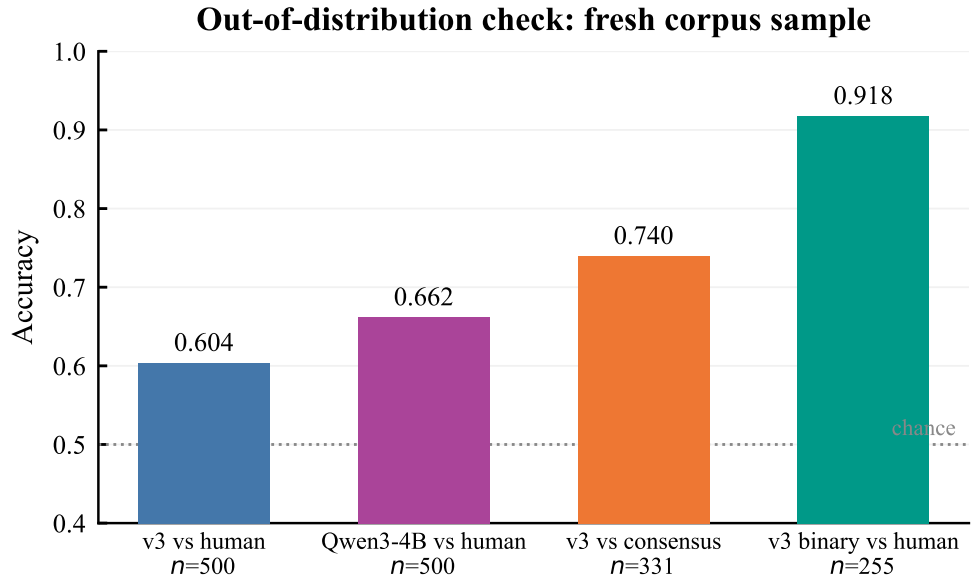


Figure 6: lab500 accuracy by evaluator/reference pair. The binary task generalises far better than three-class.

Results.

Findings.

1. **The gap is real but concentrated.** Three-class accuracy drops to 60–74%, while binary (positive vs. negative, $n = 255$) stays at 91.8% — polarity is easy; the neutral boundary is what fails.

2. **The bottleneck is annotation ambiguity, not model capacity.** Inter-annotator agreement between two independent LLMs is only 66.2%; a MacBERT model trained on 400 in-domain labels reproduces held-out labels at just 60% OOF. The corpus's terse, context-free comments (median 25 chars vs. gold's 80) are intrinsically ambiguous — the effective ceiling on this data is far below the gold benchmark.

3. **Difficulty is predictable.** Rows the teacher found easy (confidence ≥ 0.7) score 0.757 for every annotator pair; hard rows score ≈0.53. The gap is a difficulty difference, not evidence of training leakage (gold overlap is 3/600 rows, negligible).

We report this as a limitation, not a hidden flaw: it is precisely why the project invested in corpus-scale pseudo-labelling, and it marks the direction for future work (human annotation of terse comments).

9 Limitations and Threats to Validity

- **Domain narrowness.** The gold set is 600 reviews from a *single film page*; the lab500 study confirms the distribution shift is the dominant external-validity threat.
- **Pseudo-label ceiling.** Pseudo labels inherit teacher biases; the flywheel saturates after one round (round-2 mean confidence 0.986 = memorisation).
- **Statistical power.** At $n = 600$, differences below $\sim$1pp are unresolvable; corpus-scale claims rely on rating-correlation evidence, not human labels.
- **lab500 labels are LLM-assisted**, not gold-standard human annotations; the 60–74% range should be read as "agreement with available annotators", bounded below by the 66% LLM–LLM agreement itself.
- **Engineering constraints.** The ROCm/Windows GPU stack required a driver-level reboot during 1.3B feature extraction; inference latency figures are single-GPU measurements.

10 Conclusion and Future Work

On a realistic low-resource Chinese sentiment task, corpus-scale pseudo-labelling is the highest-leverage ingredient: it lifts neutral recall from 54% to 71% and lets a single compact student nearly match a multi-model ensemble. Specialist decomposition on corpus-trained neutral detectors adds a further validated gain. Star ratings alone are insufficient supervision — measured, not assumed. Equally important, our honest OOD study shows the gold benchmark overstates real-world three-class performance on terse corpus comments; binary polarity, however, generalises well (91.8%).

Future work: (i) expand human annotation of terse corpus comments to 1,000+ rows to tighten confidence intervals and resolve the neutral boundary; (ii) use stronger LLM teachers for corpus re-labelling to inject knowledge the student lacks; (iii) test MacBERT-large students and the full 271K unbalanced pool; (iv) domain-adversarial or multi-domain training to close the gold→corpus gap.

Reproducibility

All scripts, out-of-fold predictions, and experiment records are in the project repository (`https://github.com/shuakami/douban-sentiment`). The 911K corpus is external (research use only, not redistributed); trained checkpoints are published at `https://huggingface.co/Shuakami/douban-sentiment-we` The full preprocessing → pseudo-label → train → evaluate pipeline is scripted end-to-end. Key artifacts: `out/ens_lite_3class_v3.json` (deployed OOF predictions + probabilities), `out/lab500_*` (OOD study), `scripts/` (all training/eval code), `PAPER.md` (condensed report). Hardware: AMD RX 9070 GRE, ROCm, PyTorch bf16.

A Training Details and Reproducibility

A.1 Hyperparameters

Table 7: Key hyperparameters of the deployed pipeline.

Component	Setting	Value
Distilled student	backbone	`hfl/chinese-macbert-base`
	distill α/τ	0.5 / 2
	epochs / max-len	8 / 256
	folds (OOF)	5
Judge head	backbone	MegatronBERT-1.3B (frozen)
	features	4096-d (CLS + mean-pooled)
	head	2-layer MLP
Neutral detector	backbone	MacBERT-base
	train rows	115,000 (per-class cap 60K, 2 epochs)
Pseudo pool	confidence	$\max p \geq 0.7$
	rows	271,905 (56.5K/68.6K/146.8K)
Calibration	method	temperature scaling
	T	0.37 (NLL on gold)
Ensemble weights	dep	student 1 : judge 2
	spec blend	0.5
	p_n mix	0.6 judge-head : 0.4 corpus detector

A.2 Artifacts

- `out/ens_lite_3class_v3.json` — deployed 3-class OOF predictions and probabilities (all 600 rows).
- `out/ens_lite_binary.json` — deployed binary OOF.
- `out/lab500_*` — OOD study: samples, dual-LLM labels, predictions, cross-validation results.
- `scripts/` — end-to-end pipeline covering corpus preprocessing, pseudo-labelling, corpus-head training, distillation, and ensemble evaluation (full script list in the repository).
- `web/` + `web-ui/` — FastAPI backend and React demo.

A.3 Environment

Python 3.12, PyTorch (ROCm), `transformers`, AMD RX 9070 GRE GPU. All GPU jobs serialized (concurrent jobs crash the ROCm/Windows stack). Model weights: MacBERT students ∼400MB each; MegatronBERT-1.3B shared backbone; total deployed footprint under 4GB. Cold start ≈14s, warm inference ≈0.1s per review.

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